

RMC_WR_TCAM_Spec_v1

RMC — WR_TCAM Microarchitecture Specification

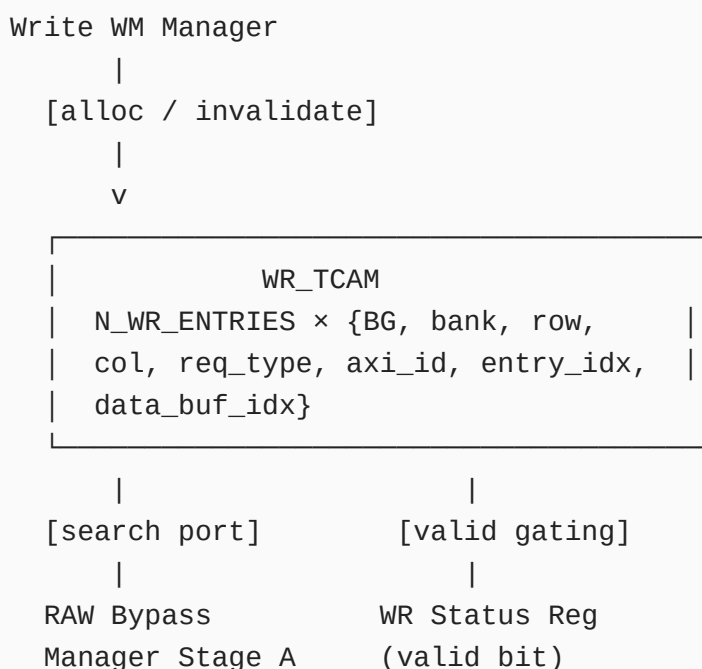
1. Overview

The WR_TCAM is a fully associative exact-match Content-Addressable Memory instantiated once in the MC Core clock domain. It stores the translated memory addresses of all outstanding write transactions and is the primary lookup structure for RAW (Read-After-Write) hazard detection.

Every cycle, Stage A of the RAW Bypass Manager issues a parallel search across all `N_WR_ENTRIES` of the WR_TCAM using an incoming read transaction's translated address as the match key. If one or more entries match, the RAW Bypass Manager determines whether the associated write data in the Write Data Buffer can fully or partially serve the read — avoiding a DRAM round-trip for data that is already pending in the write pipeline.

The WR_TCAM has no scheduling role. The Scheduler and RAW Bypass Manager have read-only (search) access. All write (allocation and invalidation) operations are owned exclusively by the Write Watermark Buffer Manager.

1.1 Position in the Pipeline



1.2 Access Rights

Accessor	Access Type
Write Watermark Buffer Manager	R/W — alloc and invalidate
RAW Bypass Manager (Stage A)	READ ONLY — search
Scheduler Stage 1	READ ONLY — search (hit bitmap only)

2. Parameters

Parameter	Width / Value	Description
N_WR_ENTRIES	64 (v1), 96 (v2)	Number of TCAM entries. Sized 1:1 with the write buffer depth.
BG_BITS	Compile-time param	Bank group address bits
BANK_BITS	Compile-time param	Bank address bits
ROW_BITS	Compile-time param	Row address bits
COL_BITS	Compile-time param	Column address bits
AXI_ID_WIDTH	Compile-time param	AXI transaction ID width
ADDR_KEY_WIDTH	BG_BITS + BANK_BITS + ROW_BITS + COL_BITS	Total match key width
ENTRY_IDX_WIDTH	$\lceil \log_2(N_WR_ENTRIES) \rceil$	Index into WR Status Reg
DATA_BUF_IDX_WIDTH	5b	Index into Write Data Buffer

All parameters are resolved at synthesis time. No hardcoded values appear in this specification.

3. Entry Structure

Each of the N_WR_ENTRIES TCAM rows stores the following fields:

Field	Width	Description
bg[BG_BITS-1:0]	BG_BITS	Bank group — exact match field
bank[BANK_BITS-1:0]	BANK_BITS	Bank — exact match field
row[ROW_BITS-1:0]	ROW_BITS	Row address — exact match field

Field	Width	Description
col[COL_BITS-1:0]	COL_BITS	Column address — exact match field
req_type	1b	Always 1b1 (WR). Hardwired at allocation.
axi_id[AXI_ID_WIDTH-1:0]	AXI_ID_WIDTH	AXI transaction ID of the write
entry_idx[ENTRY_IDX_WIDTH-1:0]	ENTRY_IDX_WIDTH	Index into WR Status Reg for valid/age lookup
data_buf_idx[4:0]	5b	Index into Write Data Buffer for payload retrieval

Total entry width = BG_BITS + BANK_BITS + ROW_BITS + COL_BITS + 1 + AXI_ID_WIDTH + ENTRY_IDX_WIDTH + 5

3.1 Valid Gating

The WR_TCAM does not store a valid bit internally. Occupancy is tracked exclusively in `WR_Status_Reg[i].valid`. A TCAM entry participates in a search only when:

```
match_enable[i] = wr_status_reg[i].valid
```

When `valid[i] == 0`, the match output for entry `i` is suppressed regardless of address content. This is the single source of truth for occupancy and enables dynamic power reduction at low occupancy — inactive entries draw no match current.

4. Internal Blocks

4.1 CAM Cell Array

The core of the WR_TCAM is a flat array of `N_WR_ENTRIES` CAM rows. Each row compares the full `ADDR_KEY_WIDTH`-bit match key against its stored `{bg, bank, row, col}` using XOR-based binary CAM cells — not subtractor-based magnitude comparators.

XOR CAM Cell Operation (per bit):

```
match_bit = ~(key_bit XOR stored_bit)
row_match[i] = AND_REDUCE(match_bit[ADDR_KEY_WIDTH-1:0]) AND
match_enable[i]
```

A row asserts `row_match[i]` only when every bit of the key matches the stored address AND

`match_enable[i]` is asserted. Single gate delay per bit; the AND reduction is the critical path within a single row.

Important: The XOR CAM cell is distinct from the subtractor comparators used in Scheduler Stage 2 (`gc - next_legal_cycle` , sign-bit check). These must not be conflated in the microarchitecture or RTL. Stage A is a true CAM; Scheduler Stage 2 is a registered `can_*` flag array.

4.2 Match Enable Array

An `N_WR_ENTRIES` -wide register that holds the current `wr_status_reg[i].valid` state for every entry. Updated every cycle by the Write WM Manager as entries are allocated and retired. Gates the CAM cell array output before forming `wr_hit_vector` .

```
match_enable[i] = wr_status_reg[i].valid    // sampled each cycle
wr_hit_vector[i] = row_match[i] AND match_enable[i]
```

4.3 Write Port (Allocation)

A single write port owned by the Write Watermark Buffer Manager. On write allocation, all fields of the target entry (`bg` , `bank` , `row` , `col` , `req_type` , `axi_id` , `entry_idx` , `data_buf_idx`) are written in a single cycle. The corresponding `WR_Status_Reg[i].valid` is set to `1` by the Write WM Manager in the same cycle, enabling the entry for future searches from the next cycle onward.

Only one entry is allocated per cycle (single write port). Simultaneous allocation and invalidation of different entries is supported — they target different physical rows.

4.4 Invalidate Port (Retirement)

A single invalidate port owned by the Write Watermark Buffer Manager. On write retirement (when `WR_Status_Reg[i].status` transitions to `DONE`), the Write WM Manager clears `WR_Status_Reg[i].valid` . Since valid gating is sourced from `WR_Status_Reg` , no explicit erase of the TCAM cell array is required — the entry is suppressed from all future searches immediately on the cycle `valid[i]` deasserts.

TCAM cell contents for the invalidated row are not explicitly cleared. They are overwritten on the next allocation to that row index.

4.5 Search Port

A single combinational search port driven by RAW Bypass Manager Stage A every cycle. The search port accepts an `ADDR_KEY_WIDTH` -bit match key and produces an `N_WR_ENTRIES` -bit hit vector in the same cycle (see Timing Notes).

```
search_key = {rd_bg, rd_bank, rd_row, rd_col}
wr_hit_vector[i] = row_match[i] AND match_enable[i]    // for all i in
parallel
```

The search port is not gated by `rd_alloc_valid` — it evaluates combinationally whenever a search key is presented. The RAW Bypass Manager is responsible for qualifying `wr_hit_vector` with `rd_alloc_valid` before acting on the result.

5. Interface Signals

All signals are in the MC Core clock domain. All intra-MC interfaces use valid-credit protocol (no combinational ready). No CDC crossings exist inside the WR_TCAM.

5.1 Inputs

Signal	Width	Source	Description
<code>alloc_en</code>	1b	Write WM Manager	Pulse: allocate a new entry this cycle
<code>alloc_idx[ENTRY_IDX_WIDTH-1:0]</code>	ENTRY_IDX_WIDTH	Write WM Manager	Target entry index for allocation
<code>alloc_bg[BG_BITS-1:0]</code>	BG_BITS	Write WM Manager	Bank group to store
<code>alloc_bank[BANK_BITS-1:0]</code>	BANK_BITS	Write WM Manager	Bank to store
<code>alloc_row[ROW_BITS-1:0]</code>	ROW_BITS	Write WM Manager	Row address to store
<code>alloc_col[COL_BITS-1:0]</code>	COL_BITS	Write WM Manager	Column address to store
<code>alloc_axi_id[AXI_ID_WIDTH-1:0]</code>	AXI_ID_WIDTH	Write WM Manager	AXI ID of allocating write
<code>alloc_data_buf_idx[4:0]</code>	5b	Write WM Manager	Write Data Buffer index for this entry
<code>match_enable[N_WR_ENTRIES-1:0]</code>	N_WR_ENTRIES	WR Status Reg	Per-entry valid gating; sampled every cycle

Signal	Width	Source	Description
search_key[ADDR_KEY_WIDTH-1:0]	ADDR_KEY_WIDTH	RAW Bypass Manager Stage A	{rd_bg, rd_bank, rd_row, rd_col} — match key

5.2 Outputs

Signal	Width	Destination	Description
wr_hit_vector[N_WR_ENTRIES-1:0]	N_WR_ENTRIES	RAW Bypass Manager Stage A	Per-entry match result, validated. Non-zero = at least one hit.
tcam_axi_id[N_WR_ENTRIES-1:0][AXI_ID_WIDTH-1:0]	N_WR_ENTRIES × AXI_ID_WIDTH	RAW Bypass Manager Stage B	AXI IDs of all entries; Stage B reads the winner's field after hit selection
tcam_entry_idx[N_WR_ENTRIES-1:0][ENTRY_IDX_WIDTH-1:0]	N_WR_ENTRIES × ENTRY_IDX_WIDTH	RAW Bypass Manager Stage B	WR Status Reg indices; used for age lookup in multi-hit tiebreak
tcam_data_buf_idx[N_WR_ENTRIES-1:0][4:0]	N_WR_ENTRIES × 5	RAW Bypass Manager Stage B	Write Data Buffer indices; used to read payload for bypass

Note on output bus width: All per-entry metadata fields are exposed as flat buses to the RAW Bypass Manager. Stage B muxes down to the winning entry after argmax selection. Exporting all entries avoids a read port inside the TCAM and keeps the critical path outside the array.

6. Functional Behaviour

6.1 Allocation Flow

1. Write WM Manager presents `alloc_en`, `alloc_idx`, and all address/metadata fields.
2. `WR_TCAM` writes `{bg, bank, row, col, req_type=1, axi_id, entry_idx, data_buf_idx}` into the physical row at `alloc_idx`.
3. Write WM Manager sets `WR_Status_Reg[alloc_idx].valid = 1` in the same cycle.
4. From the next cycle onward, `match_enable[alloc_idx]` is asserted and the entry participates in searches.

6.2 Invalidation Flow

1. Write WM Manager detects `WR_Status_Reg[i].status == DONE` for a completed write.
2. Write WM Manager clears `WR_Status_Reg[i].valid = 0`.
3. `match_enable[i]` deasserts immediately. The entry is suppressed from all searches from the next cycle onward.
4. Physical TCAM cell contents at row `i` are not cleared — stale data is harmless since `match_enable[i] == 0` prevents any match from firing.

6.3 Search Flow

1. RAW Bypass Manager Stage A presents `search_key = {rd_bg, rd_bank, rd_row, rd_col}` every cycle for any newly allocated or currently stalled read entry.
2. All `N_WR_ENTRIES` CAM rows evaluate simultaneously and combinationally.
3. `wr_hit_vector[i]` is asserted for every entry where the address matches AND `match_enable[i] == 1`.
4. `wr_hit_vector` is forwarded to RAW Bypass Manager Stage B in the same cycle.
5. Stage B handles multi-hit resolution (argmax on age via WR Status Reg) — this is outside the `WR_TCAM` boundary.

6.4 Simultaneous Alloc and Search

Allocation (write port) and search (search port) may occur in the same cycle targeting different entries — they are independent ports. If allocation and search target the same entry index in the same cycle, the search sees the pre-allocation state (the new entry's `match_enable` does not assert until the next cycle). This is safe by construction: a newly allocated write entry has `age == gc_now`, and a read allocated in the same cycle has `age == gc_now`. The RAW Bypass Manager's age gate (`wr_age <= rd_age`) admits same-cycle writes as valid hazard sources; however, since `match_enable` is sourced from `WR_Status_Reg.valid` which is written this cycle and read next cycle, the new entry is invisible for one cycle. This one-cycle window is acceptable — the read entry remains in the RD Status Reg and Stage A re-evaluates it every cycle until a match is confirmed or the read proceeds to the Scheduler.

6.5 Simultaneous Alloc and Invalidate

Allocation and invalidation may target different entry indices in the same cycle — no conflict. Allocation and invalidation of the same index in the same cycle is not possible by construction: a slot being retired (`DONE`) cannot simultaneously receive a new allocation until its `valid` bit is cleared, and the Write WM Manager enforces this ordering.

7. Timing Notes

- **Search latency:** Fully combinational. `wr_hit_vector` is available in the same cycle `search_key` is presented. This is consistent with OQ-1 in the RAW Bypass Manager spec (combinational parallel with RD Status Reg allocation is the recommended resolution).
 - **Allocation latency:** The written entry is visible to the search port from the **next cycle** onward, gated by `WR_Status_Reg[i].valid` which settles on the clock edge following `alloc_en`.
 - **Invalidation latency:** The retired entry is suppressed from the search port from the **next cycle** onward, as `match_enable[i]` deasserts on the clock edge following the `valid` clear.
 - **Critical path:** The AND reduction across `ADDR_KEY_WIDTH` XOR outputs within a single CAM row is the intra-row critical path. The priority encoder across `N_WR_ENTRIES` rows (if any — see note below) is the inter-row critical path. The `WR_TCAM` exports the full `wr_hit_vector` rather than a single winner index; multi-hit resolution (argmax on age) is performed by RAW Bypass Manager Stage B, keeping the priority encoder off the TCAM's critical path.
 - **Power:** Match enable gating suppresses toggle activity in idle CAM rows. At low occupancy (few valid entries), most rows see no switching on the match lines.
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8. Edge Cases

Scenario	Behaviour
All entries invalid (<code>match_enable == 0</code>)	<code>wr_hit_vector == 0</code> . No hit. RAW Bypass Manager passes read to Scheduler.
Single hit	<code>wr_hit_vector</code> has exactly one bit set. Stage B uses that entry directly — no argmax needed.
Multi-hit (same address, multiple writes)	<code>wr_hit_vector</code> has multiple bits set. Stage B resolves via <code>argmax(wr_status_reg[i].age)</code> . Newest write wins. Lowest-index-wins is explicitly incorrect and must not be implemented.

Scenario	Behaviour
Write at same address, arrived after read (<code>wr_age > rd_age</code>)	Entry participates in TCAM match (<code>match_enable</code> may be asserted), but Stage A age gate filters it out before <code>wr_hit_vector</code> is formed. The TCAM itself does not enforce the age condition — this is the RAW Bypass Manager's responsibility.
TCAM full (<code>N_WR_ENTRIES</code> all valid)	No structural issue. The TCAM operates normally. The Write WM Manager is responsible for backpressure upstream (via watermark logic) to prevent overflow. The TCAM has no overflow detection logic.
Simultaneous allocation and search on same index	New entry invisible for one cycle (valid gating is registered). Safe — read re-evaluates next cycle.
Retirement of an entry currently being searched	<code>match_enable[i]</code> deasserts next cycle. Current cycle's search sees the pre-retirement state. If the entry was the winning hit, Stage B will select it and the RAW Bypass Manager may fire. This is correct — the write was still <code>PENDING</code> or <code>ISSUED</code> at the time of the search; retirement (<code>DONE</code>) takes effect the following cycle.

9. Design Decisions

Decision	Rationale
XOR CAM cells (not subtractors)	Single gate delay per bit. Subtractors are reserved for Scheduler Stage 2 timing gates only.
Valid gating sourced from <code>WR_Status_Reg</code> (not internal)	Single source of truth for occupancy. Avoids synchronization hazard between TCAM valid state and status reg valid state.
Full <code>wr_hit_vector</code> exported (not winner index)	Keeps argmax (age comparator tree) outside the TCAM boundary in Stage B. Shorter TCAM critical path.
No explicit invalidation erase	Stale cell data is harmless when <code>match_enable == 0</code> . Saves an erase cycle on retirement.
Single write port	Write WM Manager issues at most one write allocation per cycle. No multi-port complexity required.
<code>col</code> included in match key	Writes to same row but different columns cover different 64B regions and are not RAW hazards to each other. Column is not a don't-care.

10. Block Summary

Write WM Manager

